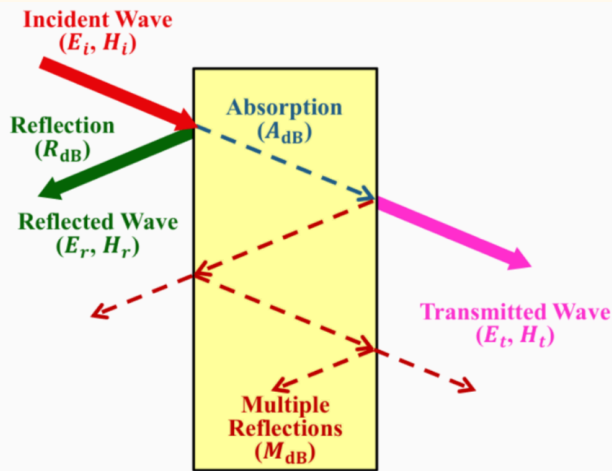


Shielding Effectiveness



$$\text{Shielding Effectiveness, dB} = SE_{dB}(f)$$

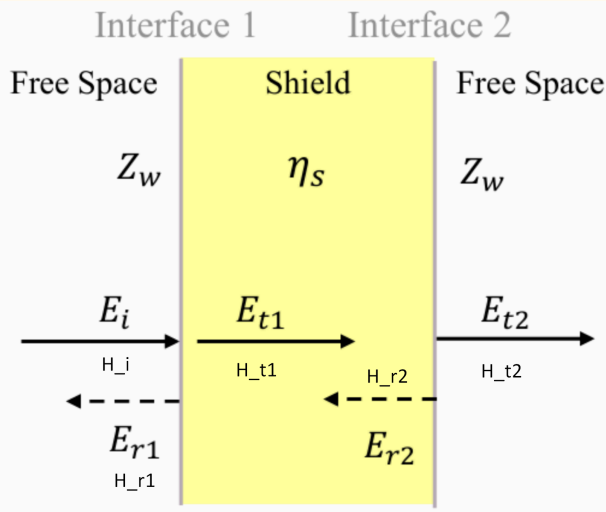
$$= 20 \log_{10} \left(\left| \frac{E_{incident}(f)}{E_{transmitted}(f)} \right| \right)$$

$$= 20 \log_{10} \left(\left| \frac{E_i(f)}{E_t(f)} \right| \right)$$

$$= 20 \log_{10} \left(\left| \frac{H_i(f)}{H_t(f)} \right| \right)$$

$$= R_{dB} + A_{dB} + M_{dB}$$

Reflection Loss



$$\text{Reflection coefficient, } \Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\text{Transmission coefficient, } \tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

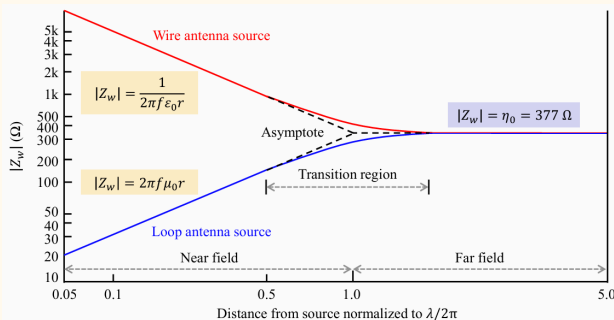
$$R_{dB}(E) = 20 \log_{10} \left(\left| \frac{E_i}{E_{t2}} \right| \right) = 20 \log_{10} \left(\left| \frac{(Z_w + \eta_s)^2}{4Z_w \eta_s} \right| \right)$$

$$R_{dB}(H) = 20 \log_{10} \left(\left| \frac{H_i}{H_{t2}} \right| \right) = 20 \log_{10} \left(\left| \frac{(Z_w + \eta_s)^2}{4Z_w \eta_s} \right| \right)$$

For good conductor $\eta_s \ll Z_w$:

$$R_{dB} \approx 20 \log_{10} \left(\left| \frac{Z_w}{4\eta_s} \right| \right)$$

Wave Impedance of Different Sources



$$\text{Reactive Near Fields: } r < r_1 = \frac{1}{2\pi} \lambda = 0.16 \lambda \quad (kr = 1)$$

$$|Z_{w, \text{wire antenna}}| = \frac{1}{2\pi f \epsilon_0 r}$$

$$|Z_{w, \text{loop antenna}}| = 2\pi f \mu_0 r$$

$$|Z_{w, \text{far field}}| = 120\pi$$

Intrinsic Impedance of Shield Material

$$\eta_{complex} = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon - j\frac{\sigma}{\omega}}} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

$$\epsilon_c = \epsilon' - j\epsilon'' = \epsilon - j\frac{\sigma}{\omega}$$

$$Loss\ tangent = \frac{\epsilon''}{\epsilon'} = \frac{\epsilon_r''}{\epsilon_r'}$$

$$Loss\ tangent = \frac{\frac{\sigma}{\omega}}{\epsilon} = \frac{\sigma}{\omega\epsilon}$$

$\sigma \gg \omega\epsilon$:

$$\eta_{GoodConductor} = \sqrt{\frac{j\omega\mu}{\sigma}} = (1 + j)\sqrt{\frac{\omega\mu}{2\sigma}}$$

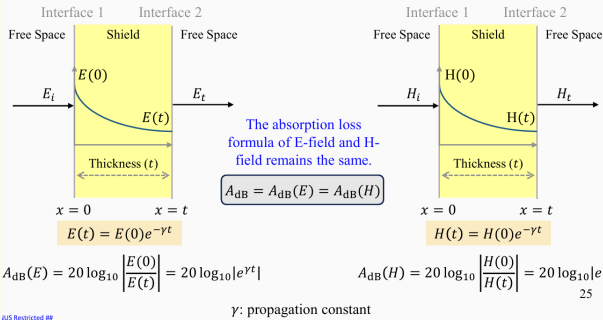
$$|\eta_{GoodConductor}| = \sqrt{\frac{\omega\mu}{\sigma}}$$

$\sigma \ll \omega\epsilon$:

$$\eta_{LowLossDielectric} = \sqrt{\frac{\mu}{\epsilon}}$$

Absorption Loss

□ E-field Case:



Attenuation constant, $\alpha = \sqrt{\pi f \mu \sigma}$

Skin depth (good conductor), $\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$

$$A_{dB}(f) = 20 \log_{10}(e^{t/\delta})$$

Multiple Reflection Loss

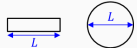
Only for thin shield (A_{dB} is small) and near field H field sources (R_{dB} is small)

Thin shield: $t < \delta$

$$M_{dB}(H) = 20 \log_{10}(1 - e^{-2t/\delta})$$

Effect of Aperture

□ For a single aperture with its dimensions not larger than 1/2 wavelength of the frequency, a rule-of-thumb formula can be used to predict the SE due the aperture shape.



$$SE_{dB} = k \log_{10} \frac{\lambda}{2L}$$

where λ is the wavelength and L is the maximum linear dimension of the aperture. $k = 20$ for rectangular apertures and $k = 40$ for circular apertures.

Multiple apertures will reduce the SE of an enclosure. For a linear array of closed spaced n apertures, a correction factor needs to be considered.

$$\Delta SE_{dB} = -20 \log_{10} \sqrt{n}$$



$$SE_{dB} = k \log_{10} \frac{\lambda}{2L} - 20 \log_{10} \sqrt{n}$$

The above formula is accurate for a liner array of apertures; however, it is still effective for a 2D array of apertures (or holes) because adding more rows or columns of apertures does not significantly compromise the SE.

Smaller holes provide better SE_{dB} , but multiple holes would reduce SE_{dB}